FOR A REPRESENTABLE MORPHISM, QUASI-SEPARATEDNESS IS EQUIVALENT TO THE DIAGONAL BEING QUASI-COMPACT AND QUASI-SEPARATED

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter source sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the Stacks Project keeps one single shared preamble for every chapter; this card takes it directly from the staged sources/stacks-algebraic-stacks-morphisms/preamble.tex as three verbatim spans (lines 8-12, 14-18 and 154-217), so no derived file was written and nothing was added or redefined. The wrapper class and the base packages are supplied by the assembler.

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Editorial note. Selection provenance: this card was opened under the file-based control contract (CONTROL.md), with the mechanical pre-card gate run on its statement and proof interval before the card was written.

Allowed standard prerequisites (editorial).

- The base change formula for the diagonal. Invoked at the start of the counted proof (source lines 510-514) as a cross-chapter pointer. Identifies the diagonal of a base change with a base change of the diagonal.
- The criterion for checking a property of a representable morphism along a smooth surjective cover. Invoked inside the counted proof (source lines 515-525) as a cross-chapter pointer. Reduces the property of the diagonal to the presenting morphism of algebraic spaces.
- The corresponding equivalence for morphisms of algebraic spaces. Invoked inside the counted proof (source lines 526-535) as a cross-chapter pointer. Gives the equivalence for the presenting morphism.

Source reference. Original reference algebraic-lemma-representable-transformations-property-implication: sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex, source lines 522–522. Navigation pointer only: the counted text refers to the Stacks item algebraic-lemma-representable-transformations-property-implication, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Source reference. Original reference categories-lemma-base-change-diagonal: sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex, source lines 476–476. Navigation pointer only: the counted text refers to the Stacks item categories-lemma-base-change-diagonal, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because

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Source reference. Original lemma lemma-properties-diagonal-representable: sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex, source lines 405–405. Navigation pointer only: the counted text refers to the Stacks item lemma-properties-diagonal-representable, which is an existing labeled result of this same chapter and is not part of this card. The pointer carries the target's own label as its number and its definition line as its span; its content is not reproduced here.

Source reference. Original reference stacks-properties-lemma-check-property-covering: sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex, source lines 479–479. Navigation pointer only: the counted text refers to the Stacks item stacks-properties-lemma-check-property-covering, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Source reference. Original reference stacks-properties-section-properties-morphisms: sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex, source lines 248–248. Navigation pointer only: the counted text refers to the Stacks item stacks-properties-section-properties-morphisms, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Lemma 0.1. *Let $f : \mathcal{X} \to \mathcal{Y}$ be a morphism of algebraic stacks representable by algebraic spaces. Then the following are equivalent*

(1) *f is quasi-separated,*
(2) *Δ_f is quasi-compact, or*
(3) *Δ_f is of finite type.*

Proof. The statements "f is quasi-separated", "Δ_f is quasi-compact", and "Δ_f is of finite type" refer to the notions defined in Properties of Stacks, Section stacks-properties-section-properties-morphisms. Note that (2) and (3) are equivalent in view of the fact that Δ_f is locally of finite type by Lemma lemma-properties-diagonal-representable (and Algebraic Stacks, Lemma algebraic-lemma-representable-transformations-property-implication). Choose a scheme V and a surjective smooth morphism $V \to \mathcal{Y}$. Set $U = \mathcal{X} \times_{\mathcal{Y}} V$ which is an algebraic space by assumption, and the morphism $U \to \mathcal{X}$ is surjective and smooth. By Categories, Lemma categories-lemma-base-change-diagonal and Properties of Stacks, Lemma stacks-properties-lemma-check-property-covering we see that we have: Δ_f is quasi-compact if and only if $\Delta_{U/V} : U \to U \times_V U$ is quasi-compact. If (1) holds, then $U \to V$ is quasi-separated and we see that $\Delta_{U/V}$ is quasi-compact, i.e., (2) holds. Assume (2) holds. Let T be a scheme, and $a : T \to \mathcal{Y}$ a morphism. Set $T' = \mathcal{X} \times_{\mathcal{Y}} T$. To prove (1) we have to show that the morphism of algebraic spaces $T' \to T$ is quasi-separated. Using Categories, Lemma categories-lemma-base-change-diagonal once more we see that $\Delta_{T'/T}$ is the base change of Δ_f. Hence our assumption (2) implies that $\Delta_{T'/T}$ is quasi-compact, hence $T' \to T$ is quasi-separated as desired. □